As both the blocks are connected to the unstretchable string, if m_1 moves with an acceleration a downward then m_2 also moves with the same acceleration a horizontally.

The forces acting on mass m_2 are

(i) Downward gravitational force (m_2g)
(ii) Upward normal force (N) exerted by the surface
(iii) Horizontal tension (T) exerted by the string

The forces acting on mass m_1 are

(i) Downward gravitational force (m_1g)
(ii) Tension (T) acting upwards

The free body diagrams for both the masses is shown in Figure 3.18.

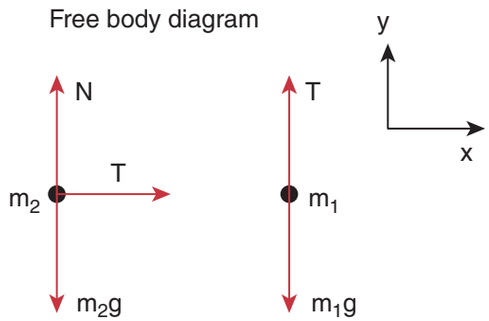


Figure 3.18 Free body diagrams of masses m_1 and m_2

Applying Newton's second law for m_1

$$T\hat{j} - m_1 g\hat{j} = -m_1 a\hat{j} \text{ (along y direction)}$$

By comparing the components on both sides of the above equation,

$$T - m_1 g = -m_1 a \qquad (3.14)$$

Applying Newton's second law for m_2

$$T\hat{i} = m_2 a\hat{i} \text{ (along x direction)}$$

By comparing the components on both sides of above equation,

$$T = m_2 a \qquad (3.15)$$

There is no acceleration along y direction for m_2.

$$N\hat{j} - m_2 g\hat{j} = 0$$

By comparing the components on both sides of the above equation

$$N - m_2 g = 0$$

$$N = m_2 g \qquad (3.16)$$

By substituting equation (3.15) in equation (3.14), we can find the tension T

$$m_2 a - m_1 g = -m_1 a$$

$$m_2 a + m_1 a = m_1 g$$

$$a = \frac{m_1}{m_1 + m_2} g \qquad (3.17)$$

Tension in the string can be obtained by substituting equation (3.17) in equation (3.15)

$$T = \frac{m_1 m_2}{m_1 + m_2} g \qquad (3.18)$$

Comparing motion in both cases, it is clear that the tension in the string for horizontal motion is half of the tension for vertical motion for same set of masses and strings.

This result has an important application in industries. The ropes used in conveyor belts (horizontal motion) work for longer duration than those of cranes and lifts (vertical motion).

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